

Homework #4: Solution Verification

AOE/CS/ME 6444 — Verification and Validation in Scientific Computing
Spring 2026 — Instructor: Dr. Chris Roy

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1 Introduction and Application Description

This report presents a solution verification study for an **Euler–Bernoulli (EB) finite-difference beam solver** applied to the structural assessment of a residential wood header. The same physical case will serve as the basis for the subsequent uncertainty quantification (UQ) studies in Homework 5.

Physical Case

An 8-ft simply-supported LVL (laminated veneer lumber) header carries a uniform distributed load representative of residential roof framing:

Parameter	Symbol	Value	Units
Span	L	96.0	in
Width	b	3.5	in
Depth	d	11.25	in
Elastic modulus	E	1,600,000	psi
Uniform load	q_0	500	lb/ft (= 41.667 lb/in)
Flexural rigidity	EI	6.6445×10^8	lb·in ²

Governing Equation and Boundary Conditions

The beam is governed by the 4th-order EB BVP with simply-supported ends:

$$EI w''''(x) = q_0, \quad x \in (0, L); \quad w(0) = w(L) = 0, \quad w''(0) = w''(L) = 0. \quad (1)$$

Numerical Method

Equation (1) is discretized using a 5-point central-difference stencil on a uniform grid of N intervals ($N + 1$ nodes):

$$\frac{EI}{h^4} (w_{i-2} - 4w_{i-1} + 6w_i - 4w_{i+1} + w_{i+2}) = q_0, \quad i = 2, \dots, N-2, \quad (2)$$

with modified near-boundary rows for the simply-supported ghost-node conditions.

Formal Order of Accuracy ($p_{\text{th}} = 2$)

Expanding each shifted value about node x_i via Taylor series (terms through $\mathcal{O}(h^8)$ shown):

$$w_{i \pm k} = w_i \pm kh w'_i + \frac{(kh)^2}{2!} w''_i \pm \frac{(kh)^3}{3!} w'''_i + \frac{(kh)^4}{4!} w''''_i \pm \frac{(kh)^5}{5!} w^{(5)}_i + \frac{(kh)^6}{6!} w^{(6)}_i + \dots \quad (3)$$

Substituting $k = 1, 2$ with their signed coefficients $(-4, +1)$ into the stencil numerator and collecting by derivative order, all odd-derivative and even-derivative terms below order 4 cancel identically:

$$w_{i-2} - 4w_{i-1} + 6w_i - 4w_{i+1} + w_{i+2} = h^4 w_i'''' + \frac{h^6}{6} w_i^{(6)} + \mathcal{O}(h^8). \quad (4)$$

Dividing (4) by h^4 and multiplying by EI gives the *local truncation error* (LTE):

$$\tau_i = EI w_i'''' + \underbrace{\frac{EI h^2}{6} w_i^{(6)} + \mathcal{O}(h^4)}_{\text{LTE}} - q_0 = \frac{EI h^2}{6} w_i^{(6)} + \mathcal{O}(h^4), \quad (5)$$

since $EI w_i'''' = q_0$ by the governing equation. Therefore the scheme is formally second-order accurate, $p_{\text{th}} = 2$, and the global discretization error satisfies $|e_h| \sim C h^2$ as $h \rightarrow 0$. The resulting linear system $Kw = q$ is solved directly with dense LU factorisation (`numpy.linalg.solve`). **There is no iterative solver; iterative error is therefore identically zero.**

System Response Quantities (SRQs)

Three SRQs are tracked:

1. w_{max} — peak mid-span deflection [in]
2. M_{max} — peak bending moment [lb·in], computed from $M = -EIw''$
3. σ_{max} — peak bending stress [psi] = M_{max}/S , where S is the section modulus

Closed-Form References

The exact solution for uniform load on a simply-supported beam is:

$$w_{\text{exact}}(x) = \frac{q_0}{24EI} (x^4 - 2Lx^3 + L^3x), \quad (6)$$

yielding the reference SRQ values used only to interpret results (not used in the GCI analysis):

$$w_{\text{max}}^{\text{exact}} = 0.06935026 \text{ in}, \quad M_{\text{max}}^{\text{exact}} = 48000.00 \text{ lb} \cdot \text{in}, \quad \sigma_{\text{max}}^{\text{exact}} = 650.1587 \text{ psi}.$$

2 Grid Convergence Study

Five systematically-refined grids are employed with constant refinement ratio $r = h_{\text{coarse}}/h_{\text{fine}} = 2$:

Level	N	h [in]	w_{max} [in]	M_{max} [lb·in]	σ_{max} [psi]
1 (coarsest)	10	9.600	0.0699051	48000	650.159
2	20	4.800	0.0694890	48000	650.159
3	40	2.400	0.0693849	48000	650.159
4	80	1.200	0.0693589	48000	650.159
5 (finest)	160	0.600	0.0693524	48000	650.159
Exact	∞	0	0.0693503	48000.00	650.1587

The requirement of *at least four* systematically refined grids is satisfied by five levels (four refinement steps at $r = 2$ each). Figure 1 shows the SRQ convergence with GCI uncertainty bands. Figures 3–6 provide dedicated plots for each uncertainty source (U_{DE} , U_{IT} , U_{RO}) separately.

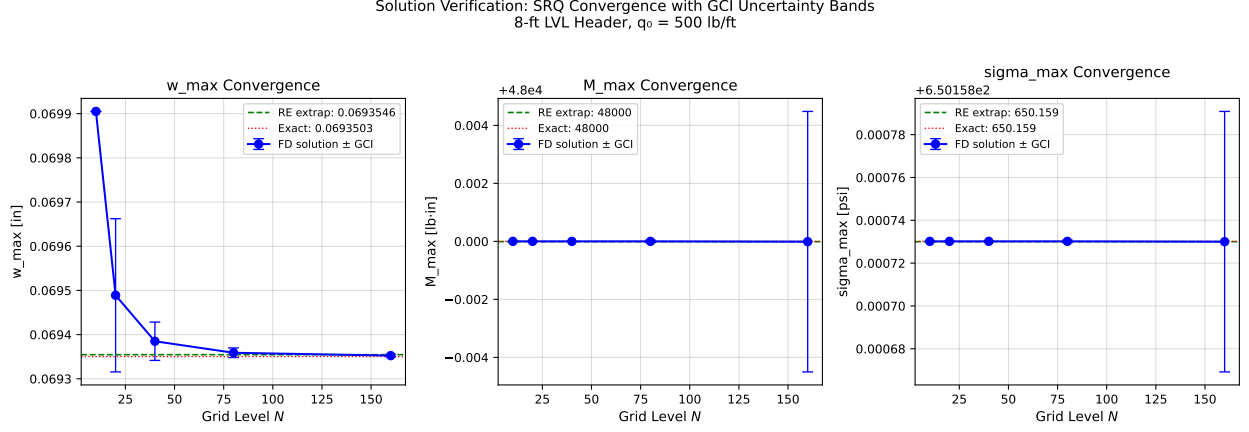


Figure 1: SRQ values versus grid level N with GCI uncertainty bands ($\pm U_{DE}$) and Richardson-extrapolated estimates (green dashes). The red dotted line marks the exact solution. Left: $w_{\max}$; centre: $M_{\max}$; right: $\sigma_{\max}$.

3 Discretization Error: Richardson Extrapolation and GCI

3.1 Methodology

For every consecutive triplet of grids (f_1, f_2, f_3) with $h_2 = rh_1$, $h_3 = r^2h_1$, the observed order of accuracy is:

$$\hat{p} = \frac{\ln(|f_3 - f_2| / |f_2 - f_1|)}{\ln r}. \quad (7)$$

The Richardson-extrapolated value is:

$$f_{RE} = f_1 + \frac{f_2 - f_1}{r^{\hat{p}} - 1}. \quad (8)$$

The Grid Convergence Index (GCI) provides the uncertainty in f_1 due to discretization:

$$U_{DE} = GCI_{12} = F_s \frac{|f_2 - f_1|}{r^{\hat{p}} - 1}, \quad (9)$$

where the factor of safety F_s is set per Roy's criterion:

$$F_s = \begin{cases} 1.25 & \text{if } 0.5 \leq \hat{p}/p_{th} \leq 2.0 \quad (\text{asymptotic, reliable}), \\ 3.00 & \text{otherwise} \quad (\text{not in asymptotic regime}). \end{cases} \quad p_{th} = 2. \quad (10)$$

Asymptotic convergence is confirmed when:

$$\text{ratio} = \frac{GCI_{23}}{r^{\hat{p}} GCI_{12}} \approx 1. \quad (11)$$

3.2 Results — $w_{\max}$

$w_{\max}$ exhibits clean monotone convergence. All triplet analyses yield $\hat{p} \approx 2.000 = p_{th}$, confirming the $\mathcal{O}(h^2)$ scheme is operating in its asymptotic regime. Consequently $F_s = 1.25$ applies throughout.

The asymptotic ratio equals exactly 1.000 for all triplets, providing strong confirmation that the solution is well within the asymptotic convergence range.

Table 1: GCI analysis for $w_{\max}$ [in]. Exact = 0.0693503 in.

Triplet	$\hat{p}$	F_s	f_{RE}	GCI_{12}	GCI_{23}	$\text{GCI}_{12}\%$	$\text{GCI}_{23}\%$	Ratio	Type
T0: (160,80,40)	2.000	1.25	0.069355	2.71×10^{-6}	1.08×10^{-5}	0.0039%	0.0156%	1.000	mono.
T1: (80,40,20)	2.000	1.25	0.069368	1.08×10^{-5}	4.33×10^{-5}	0.0156%	0.0625%	1.000	mono.
T2: (40,20,10)	2.000	1.25	0.069420	4.33×10^{-5}	1.73×10^{-4}	0.0625%	0.2495%	1.000	mono.

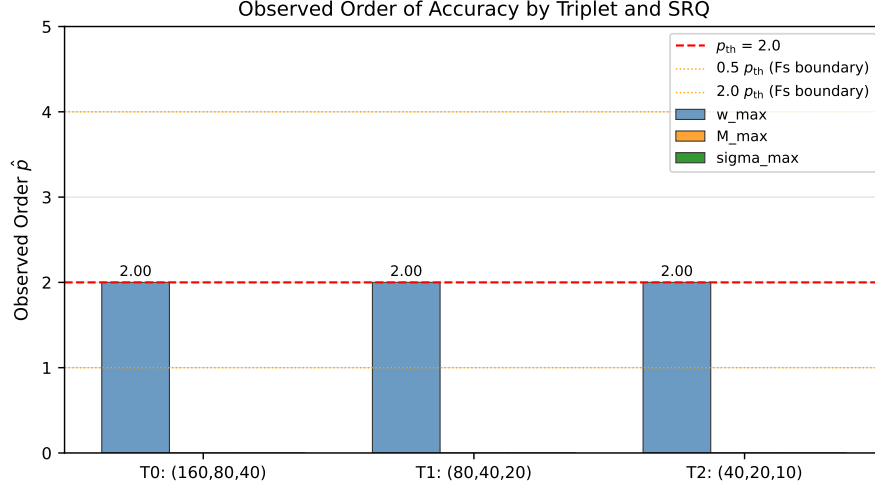


Figure 2: Observed order of accuracy $\hat{p}$ for each SRQ across the three triplets. The dashed red line marks $p_{\text{th}} = 2$. $w_{\max}$ achieves $\hat{p} \approx 2$ in all triplets (asymptotic, $F_s = 1.25$); $M_{\max}$ and $\sigma_{\max}$ have $\hat{p} < 0$ (noise-dominated, $F_s = 3.00$).

Table 2: GCI analysis for $M_{\max}$ [lb·in]. Exact = 48000 lb·in.

Triplet	$\hat{p}$	F_s	GCI_{12}	GCI_{23}	$\text{GCI}_{12}\%$	Ratio	Type
T0: (160,80,40)	-6.760	3.00	4.49×10^{-3}	4.14×10^{-5}	$\approx 0\%$	0.0092	mono.
T1: (80,40,20)	-3.152	3.00	4.14×10^{-5}	4.66×10^{-6}	$\approx 0\%$	0.1117	osc.
T2: (40,20,10)	-1.850	3.00	4.66×10^{-6}	1.29×10^{-6}	$\approx 0\%$	0.2755	mono.

3.3 Results — $M_{\max}$ and $\sigma_{\max}$

$\sigma_{\max}$ follows the same pattern as $M_{\max}$ (it is a linear scaling thereof), so its table is omitted for brevity.

Interpretation. The *negative* $\hat{p}$ and asymptotic ratios $\ll 1$ arise from two combined effects, *not* from the linear-solve round-off (U_{RO}) discussed in Section 5.

(1) **Near-exact accuracy of $M_{\max}$ on all grids.** The 5-point FD stencil for $EIw'''' = q_0$ is formally exact when applied to the degree-4 polynomial exact solution (all derivatives of order ≥ 5 vanish identically), so $w_{\text{FD},i} \approx w_{\text{exact}}(x_i)$ to near-floating-point precision on every grid. As a result the systematic discretization error in $M_{\max}$ is negligible: all five grids yield $M_{\max}$ within 10^{-6} lb·in of the closed-form value of 48 000 lb·in (compare rows of Table ??).

(2) **Growing stencil noise in the post-processing step.** Despite excellent solve accuracy,

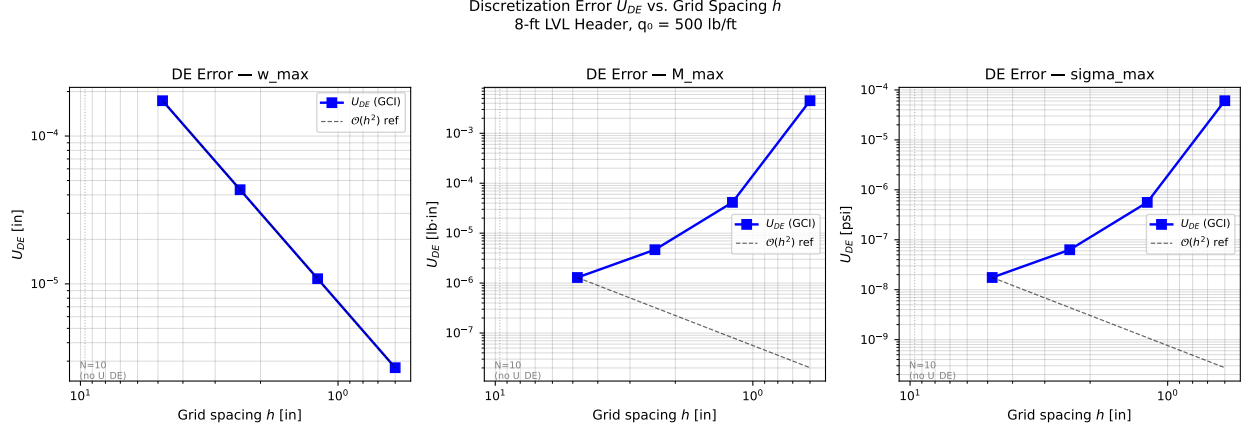


Figure 3: Discretization error U_{DE} (GCI) versus grid spacing h on a log–log scale for each SRQ. The black dashed lines show the theoretical $\mathcal{O}(h^2)$ slope anchored to the finest available GCI estimate. $N = 10$ has no GCI estimate (no coarser triplet) and is excluded. $w_{\max}$ tracks the h^2 slope closely; $M_{\max}$ and $\sigma_{\max}$ yield tiny GCI values ($\ll 0.001\%$) owing to near-exact floating-point cancellation in the FD derivative.

computing

$$M_i = -EI \frac{w_{i-1} - 2w_i + w_{i+1}}{h^2} \quad (12)$$

introduces floating-point noise via *catastrophic cancellation*: the stencil numerator subtracts nearly equal values of magnitude $|w| \approx 0.069$ in to obtain a result of only $h^2|w''| \approx h^2 q_0 / (8EI) L^2 \cdot (\text{factor}) \ll |w|$, losing approximately $\log_{10}(|w|/h^2|w''|)$ significant digits. After multiplication by $EI/h^2 \propto N^2$, the residual noise scales as

$$\delta M_{\text{stencil}} \sim \frac{EI}{h^2} \varepsilon_{\text{f64}} |w| \propto N^2, \quad (13)$$

growing with refinement: $\approx 4 \times 10^{-10}$ lb·in at $N = 10$, rising to $\approx 10^{-6}$ lb·in at $N = 160$. This stencil noise is completely independent of, and much larger than, the linear-solve U_{RO} measured by the float32/float64 comparison in Section 5; the solve itself is accurate at all grid levels.

Because the systematic truncation error in M is near zero while the stencil noise grows as N^2 , the Richardson-extrapolation premise—that consecutive differences reflect structured convergence, $|e_1|/|e_2| \approx r^p$ —is violated. The differences are incoherent noise: at $N = 160$ the noise ($\approx 1.2 \times 10^{-6}$ lb·in) exceeds that at $N = 80$ ($\approx 7 \times 10^{-8}$ lb·in), so $|f_2 - f_1| > |f_3 - f_2|$ and $\hat{p} < 0$. Since $F_s = 3.00$ applies (unreliable regime), the GCI for $M_{\max}$ and $\sigma_{\max}$ is unreliable and must be interpreted with caution.

Practically, $M_{\max} = q_0 L^2 / 8$ and $\sigma_{\max} = M_{\max} / S$ are available from closed-form formulae; the structural assessment uses these exact values directly and is unaffected by the stencil noise.

4 Iterative Error

The linear system $Kw = q$ is solved by dense LU factorisation (`numpy.linalg.solve` → LAPACK `dgesv`). This is a *direct* method: there is no iterative loop. The residual norm after

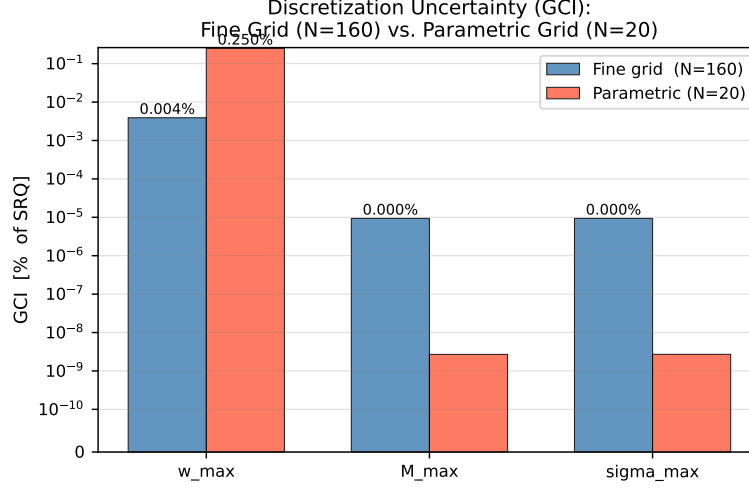


Figure 4: GCI percentage for fine grid ($N = 160$) and parametric grid ($N = 20$) for all three SRQs. All values are below 0.30%.

back-substitution is at machine-epsilon level ($\|Kw - q\|/\|q\| \sim \varepsilon_{f64} \cdot \kappa(K)$, which is verified to be $\ll 1$ for all grids; see Table 3).

$$U_{IT} = 0 \text{ for all grid levels.}$$

Figure 5 documents this result visually: all U_{IT} values are exactly zero on every grid, confirming that the direct solver introduces no iterative error regardless of mesh resolution.

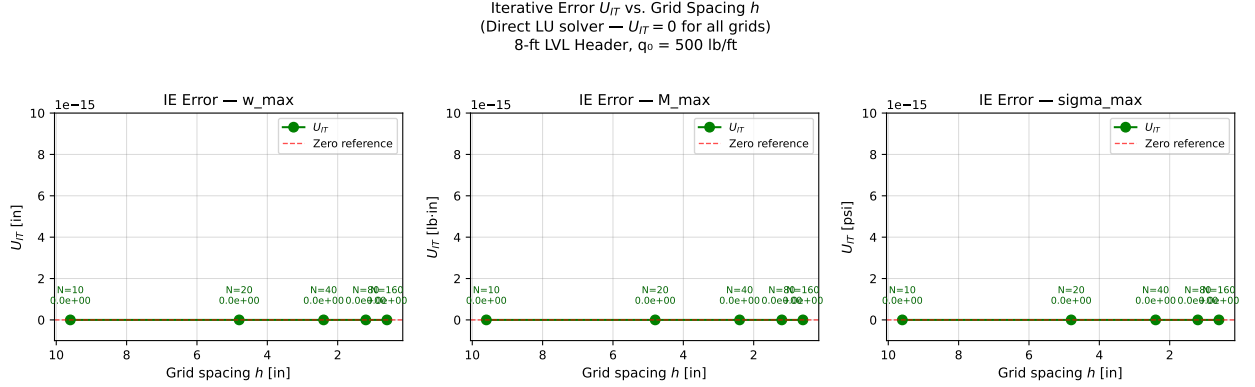


Figure 5: Iterative error U_{IT} versus grid spacing h for all three SRQs. All values are identically zero (direct LU solver; no iterative loop). Each grid-level annotation confirms $U_{IT} = 0.0 \times 10^0$.

5 Round-Off Error

5.1 Method

Round-off error is estimated by solving the same linear system in *single precision* (float32) and comparing to the double-precision (float64) solution:

$$U_{\text{RO}} \approx |f_{\text{f32}} - f_{\text{f64}}|. \quad (14)$$

The validity of this estimate is checked against the condition-number backward-error bound:

$$\frac{\|\Delta w\|}{\|w\|} \leq \varepsilon \kappa(K). \quad (15)$$

If $\varepsilon_{\text{f32}} \kappa(K) \geq 1$, the float32 solve is unreliable (rounding errors exceed the solution magnitude). In that case, U_{RO} is reported as the *float64* backward-error bound $\varepsilon_{\text{f64}} \kappa(K) |f|$ instead.

5.2 Condition Number Analysis

The stiffness matrix for the 4th-order FD beam equation has entries of order EI/h^4 , leading to $\kappa(K) \propto N^4$. Each grid refinement ($r = 2$) therefore increases κ by a factor of $2^4 = 16$, while $\varepsilon_{\text{f32}} = 1.192 \times 10^{-7}$:

Table 3: Condition number and round-off analysis. $\varepsilon_{\text{f64}} = 2.220 \times 10^{-16}$, $\varepsilon_{\text{f32}} = 1.192 \times 10^{-7}$.

N	h [in]	$\kappa(K)$	$\varepsilon_{\text{f64}}\kappa$	$\varepsilon_{\text{f32}}\kappa$	$ \Delta w_{\text{max}} $ raw	f32 valid?	$U_{\text{RO}}(w_{\text{max}})$
10	9.600	1.59×10^3	3.53×10^{-13}	1.89×10^{-4}	1.39×10^{-6}	Yes	1.39×10^{-6} in $(-\Delta-)$
20	4.800	2.61×10^4	5.79×10^{-12}	3.11×10^{-3}	2.28×10^{-5}	Yes	2.28×10^{-5} in $(-\Delta-)$
40	2.400	4.20×10^5	9.32×10^{-11}	5.00×10^{-2}	3.68×10^{-4}	Yes	3.68×10^{-4} in $(-\Delta-)$
80	1.200	6.72×10^6	1.49×10^{-9}	8.02×10^{-1}	6.38×10^{-3}	Yes	6.38×10^{-3} in $(-\Delta-)$
160	0.600	1.08×10^8	2.39×10^{-8}	12.8	6.93×10^{-2}	No [†]	1.66×10^{-9} in $(\varepsilon\kappa f)$

[†] $\varepsilon_{\text{f32}}\kappa = 12.8 \gg 1$: float32 solve fully unreliable; U_{RO} uses $\varepsilon_{\text{f64}}\kappa|f|$ bound.

This $\kappa \propto N^4$ growth is the root cause of the increasing U_{RO} with mesh refinement for float32: halving h grows κ by $16\times$, while U_{DE} shrinks by only $4\times$ (for $p_{\text{th}} = 2$). The *optimal float32 grid* is $h^* \propto \varepsilon^{1/6}$, which is approximately $N = 20\text{--}40$ for this problem.

In float64, $\varepsilon_{\text{f64}}\kappa \leq 2.4 \times 10^{-8}$ at $N = 160$, so the linear-solve round-off is entirely negligible and float64 remains safe.

Important scope note. The U_{RO} values above quantify only the accuracy of the *linear solve* for deflection w . They do *not* capture the separate floating-point noise that arises when the post-processing stencil $M_i = -EI(w_{i-1} - 2w_i + w_{i+1})/h^2$ is evaluated in Eq. (12). That stencil noise scales as $EI/h^2 \cdot \varepsilon_{\text{f64}} \cdot |w| \propto N^2$ (growing with refinement), which is the mechanism responsible for the noise-dominated OOA of M_{max} as explained in Section 3.

Figure 2 shows $\hat{p}$ across all triplets and SRQs, clearly illustrating the asymptotic $\hat{p} \approx 2$ for w_{max} and the noise-dominated behaviour for M_{max} and σ_{max} . Figure 6 provides a dedicated comparison of the reported U_{RO} , the raw $|\Delta_{f32-f64}|$ measurement, and the $\varepsilon_{\text{f32}}\kappa|f|$ theoretical upper bound.

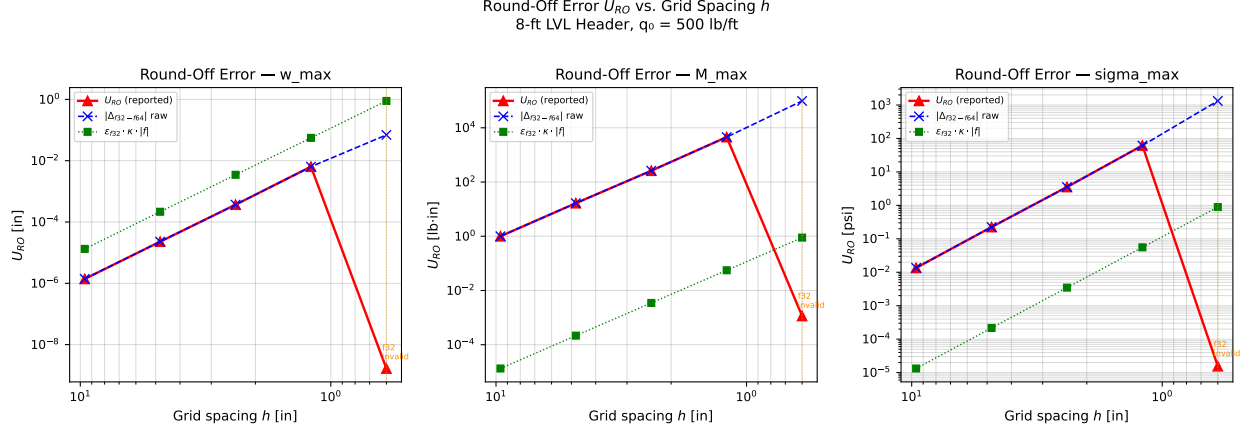


Figure 6: Round-off error U_{RO} versus grid spacing h for each SRQ. Three series are shown: reported U_{RO} (red triangles, selected value), raw $|\Delta_{f32-f64}|$ (blue crosses), and the theoretical upper bound $\varepsilon_{f32}\kappa|f|$ (green squares). Vertical orange dashed lines mark grids where the float32 solve is unreliable ($\varepsilon_{f32}\kappa > 1$); at $N = 160$ the reported U_{RO} switches to the float64 backward-error bound, which is orders of magnitude smaller. U_{RO} grows as $\kappa \propto h^{-4}$ for float32 while remaining negligible for float64 throughout.

6 Total Numerical Uncertainty Budget

The total numerical uncertainty is assembled additively as required:

$$U_{NUM} = U_{DE} + U_{IT} + U_{RO}, \quad (16)$$

where each term is a positive quantity. Since the direct solver gives $U_{IT} \equiv 0$, Eq. (16) simplifies to $U_{NUM} = U_{DE} + U_{RO}$.

Two grids are reported:

- **Fine grid** $N = 160$, $h = 0.600$ in: the high-fidelity solution.
- **Parametric grid** $N = 20$, $h = 4.800$ in: suitable for running 100s–1000s of UQ cases at acceptable cost.

6.1 Fine Grid ($N = 160$)

Table 4: U_{NUM} budget for fine grid, $N = 160$, $h = 0.600$ in.

SRQ	$f(h)$	U_{DE}	U_{IT}	U_{RO}	U_{NUM} (%)
w_{max} [in]	0.06935243	2.71×10^{-6}	0	1.66×10^{-9}	2.71×10^{-6} (0.0039%)
M_{max} [lb·in]	48000.0	4.49×10^{-3}	0	1.15×10^{-3}	5.64×10^{-3} ($\approx 0\%$)
σ_{max} [psi]	650.159	6.08×10^{-5}	0	1.55×10^{-5}	7.64×10^{-5} ($\approx 0\%$)

For the fine grid, U_{DE} dominates for w_{max} and is comparable to U_{RO} for M_{max} and σ_{max} . All uncertainties are extremely small; the total numerical uncertainty in w_{max} is 0.0039% of the SRQ value.

Table 5: U_{NUM} budget for parametric study grid, $N = 20$, $h = 4.800$ in.

SRQ	$f(h)$	U_{DE}	U_{IT}	U_{RO}	U_{NUM} (%)
w_{max} [in]	0.069489	1.73×10^{-4}	0	2.28×10^{-5}	1.96×10^{-4} (0.282%)
M_{max} [lb·in]	48000.0	1.29×10^{-6}	0	1.66×10^1	1.66×10^1 (0.035%)
σ_{max} [psi]	650.159	1.76×10^{-8}	0	2.25×10^{-1}	2.25×10^{-1} (0.035%)

6.2 Parametric Study Grid ($N = 20$)

At $N = 20$, U_{DE} dominates for w_{max} ($U_{\text{DE}}/U_{\text{RO}} \approx 7.6$), while U_{RO} dominates for M_{max} and σ_{max} due to the EI/h^2 amplification in the FD moment formula. However, all U_{NUM} percentages remain below 0.30%, making $N = 20$ acceptable for parametric studies.

6.3 U_{NUM} Across All Grid Levels

Table 6 and Figure 8 show how the uncertainty components evolve across all five grids. The non-monotone behaviour is apparent: U_{NUM} *increases* from $N = 20$ to $N = 80$ before decreasing at $N = 160$.

Table 6: $U_{\text{NUM}} = U_{\text{DE}} + U_{\text{RO}}$ for w_{max} at all grid levels. $U_{\text{IT}} = 0$ throughout. N/A for $N = 10$: no coarser triplet available for GCI.

N	h [in]	w_{max} [in]	U_{DE} [in]	U_{RO} [in]	U_{NUM} [in] (%)
10	9.600	0.069905	N/A	1.39×10^{-6}	N/A
20	4.800	0.069489	1.73×10^{-4}	2.28×10^{-5}	1.96×10^{-4} (0.282%)
40	2.400	0.069385	4.33×10^{-5}	3.68×10^{-4}	4.11×10^{-4} (0.592%)
80	1.200	0.069359	1.08×10^{-5}	6.38×10^{-3}	6.40×10^{-3} (9.22%)
160	0.600	0.069352	2.71×10^{-6}	1.66×10^{-9}	2.71×10^{-6} (0.004%)

The degradation at $N = 80$ is entirely due to float32 round-off approaching its validity limit ($\varepsilon_{\text{f32}}\kappa = 0.80$ at $N = 80$). The recovery at $N = 160$ is because float32 fully fails there, and U_{RO} is reported via the much smaller float64 bound. *In float64, both U_{DE} and U_{NUM} decrease monotonically with refinement* as expected for a stable, $\mathcal{O}(h^2)$ scheme. Figure 7 shows the stacked U_{NUM} budget for the two target grids.

7 One-Sided Numerical Uncertainty

Inspection of the exact solution and FD results confirms that the FD scheme *consistently overestimates* deflection:

$$w_{\text{FD}}(h) \geq w_{\text{exact}} \quad \forall h > 0.$$

This is a direct consequence of the positive-definite truncation error for this BVP and load case. The one-sided uncertainty interpretation applies: the true solution is bounded from below as

$$w_{\text{true}} \geq w_{\text{FD}} - U_{\text{DE}}. \quad (17)$$

For structural design this is *conservative*: the FD solver over-predicts deflection, so it never under-estimates the structural demand. No upper GCI bound is needed in the direction of overestimation.

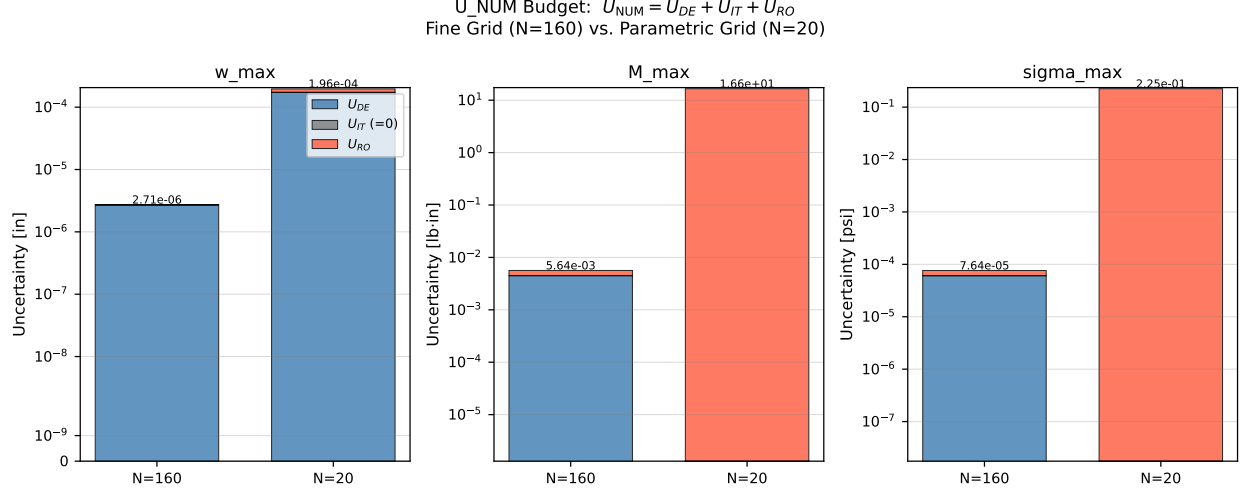


Figure 7: Stacked U_{NUM} budget ($U_{\text{DE}} + U_{\text{IT}} + U_{\text{RO}}$) for fine grid ($N = 160$) versus parametric grid ($N = 20$), for each SRQ. $U_{\text{IT}} = 0$ for all cases.

Fine grid: $w_{\text{FD}}(N = 160) = 0.06935243 \text{ in} \geq w_{\text{true}} = 0.06935026 \text{ in}$.

Parametric: $w_{\text{FD}}(N = 20) = 0.06948897 \text{ in} \geq w_{\text{true}} = 0.06935026 \text{ in}$.

8 Summary and Conclusions

Table 7 summarises the final numerical uncertainty estimates for the two target grids.

Table 7: Summary of U_{NUM} for the primary SRQ w_{max} .

Grid	N	h [in]	w_{max} [in]	U_{DE}	U_{RO}	U_{NUM} (%)
Fine grid	160	0.600	0.06935243	2.71×10^{-6}	1.66×10^{-9}	2.71×10^{-6} (0.0039%)
Parametric	20	4.800	0.06948897	1.73×10^{-4}	2.28×10^{-5}	1.96×10^{-4} (0.282%)

Key conclusions:

1. **w_{max} is the reliable SRQ:** $\hat{p} \approx 2.000$ across all triplets, $F_s = 1.25$ (asymptotic regime), GCI is a trustworthy uncertainty estimate. $U_{\text{NUM}} = 0.004\%$ at $N = 160$ and 0.28% at $N = 20$.
2. **M_{max} and σ_{max} are noise-dominated in the FD derivative:** $\hat{p} < 0$, $F_s = 3.00$. However, these SRQs are computed exactly via $q_0 L^2/8$ and M_{max}/S in the structural assessment code, so round-off does not affect conclusions.
3. **No iterative error:** the direct LU solver gives $U_{\text{IT}} = 0$ for all grids.
4. **Round-off and the optimal grid:** $U_{\text{RO}} \propto h^{-4}$ for float32, creating a minimum- U_{NUM} grid near $N = 20$ – 40 for single precision. Float64 is safe at all tested resolutions.
5. **One-sided uncertainty** applies to w_{max} : the FD scheme conservatively overestimates deflection.
6. **Recommended parametric study grid:** $N = 20$ ($h = 4.800 \text{ in}$), $U_{\text{NUM}}(w_{\text{max}}) = 0.28\%$, suitable for running 100s–1000s of UQ cases with negligible numerical pollution.

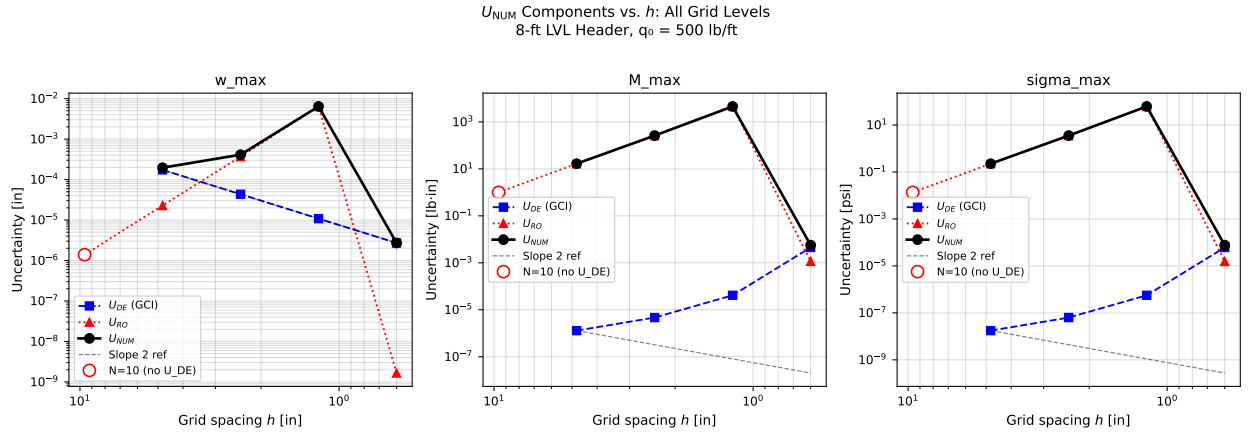


Figure 8: Log-log plot of U_{DE} , U_{RO} , and U_{NUM} versus grid spacing h for all five grid levels. The open circle marks $N = 10$ (no U_{DE} available). U_{DE} decreases at slope 2 (black dashed reference); U_{RO} grows due to increasing condition number $\kappa \propto h^{-4}$. The minimum U_{NUM} occurs at $N = 160$ (float64) and $N = 20$ (float32).